ASSIGNMENT-15.2

Backend API Development: Creating RESTful services with AI

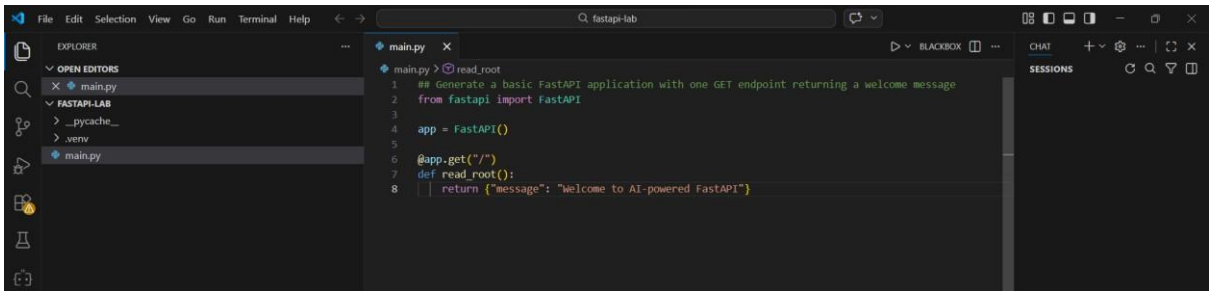
Santhosh Kumar

2403a51l21

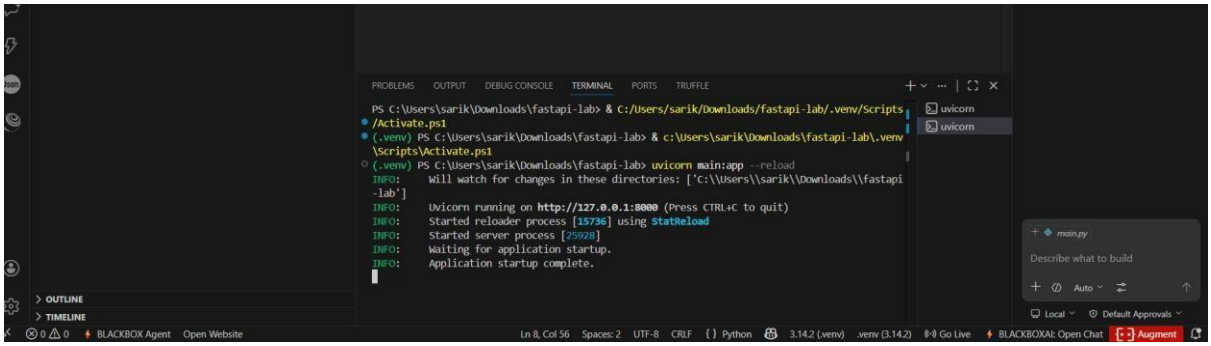
B-51

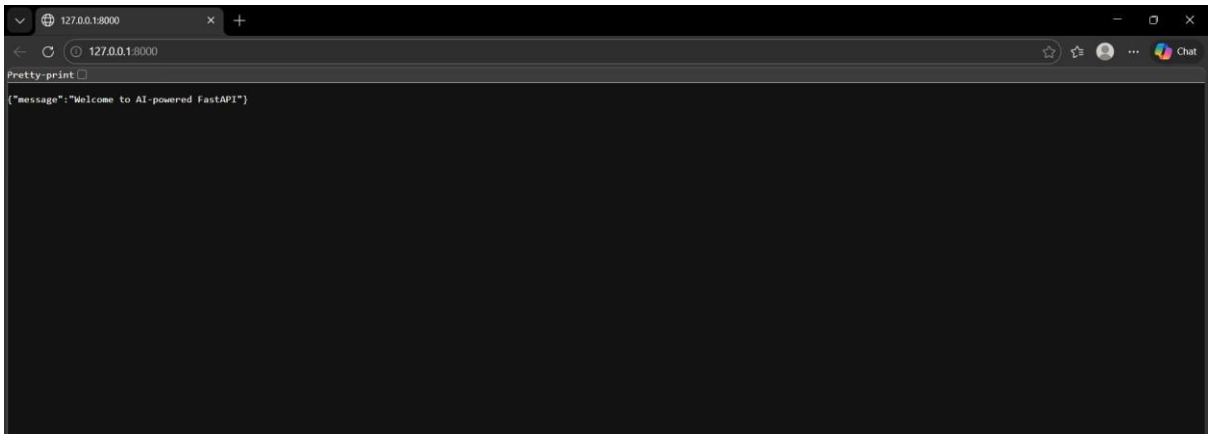
Task Descriptor-1: (Setup Fast API Backend)

PROMPT: Generate a basic FastAPI application with one GET endpoint returning a welcome message **INPUT:**



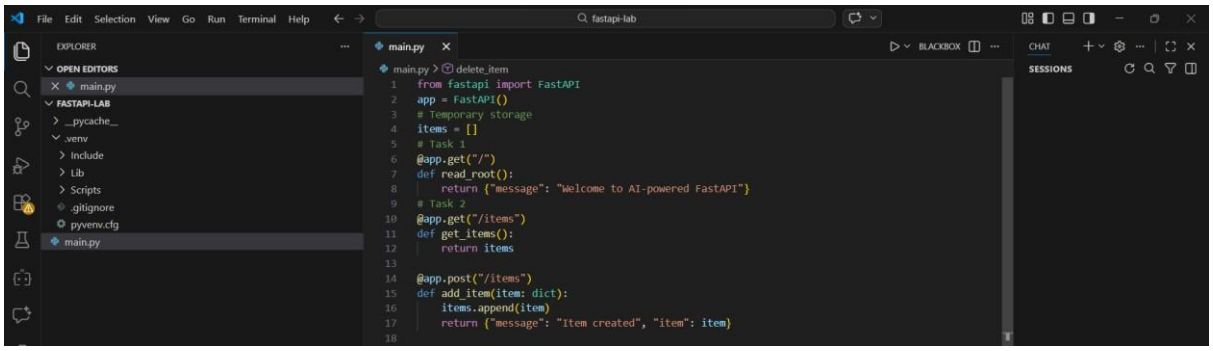
OUTPUT:





Explanation: FastAPI uses decorators to define API routes easily. The root endpoint returns a JSON response when accessed. **Task Descriptor- 2: (API – Create and Read Items)**

PROMPT: Create FastAPI endpoints to add and retrieve items using a Python list **INPUT:**



OUTPUT:

```
.gitignore
pyvenv.cfg
main.py

10  @app.get("/items")
11  def get_items():
12      return items

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   TRUFFLE        uvicorn

PS C:\Users\sarik\Downloads\fastapi-lab> & C:/Users/sarik/Downloads/fastapi-lab/.venv/Scripts/Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> pip install fastapi uvicorn
Requirement already satisfied: fastapi in .\.venv\Lib\site-packages (0.135.1)
Requirement already satisfied: uvicorn in .\.venv\Lib\site-packages (0.42.0)
Requirement already satisfied: starlette>=0.46.0 in .\.venv\Lib\site-packages (from fastapi) (0.52.1)
Requirement already satisfied: pydantic>=2.7.0 in .\.venv\Lib\site-packages (from fastapi) (2.12.5)
Requirement already satisfied: typing-extensions>=4.8.0 in .\.venv\Lib\site-packages (from fastapi) (4.15.0)
Requirement already satisfied: typing-inspection>=0.4.2 in .\.venv\Lib\site-packages (from fastapi) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in .\.venv\Lib\site-packages (from fastapi) (0.0.4)
Requirement already satisfied: click>=7.0 in .\.venv\Lib\site-packages (from uvicorn) (8.3.1)
Requirement already satisfied: h11>=0.8 in .\.venv\Lib\site-packages (from uvicorn) (0.16.0)
Requirement already satisfied: colorama in .\.venv\Lib\site-packages (from click>=7.0->uvicorn) (0.4.6)
Requirement already satisfied: annotated-types>=0.6.0 in .\.venv\Lib\site-packages (from pydantic>=2.7.0->fast
api) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.5 in .\.venv\Lib\site-packages (from pydantic>=2.7.0->fasta
pi) (2.41.5)
Requirement already satisfied: anyio<5,>=3.6.2 in .\.venv\Lib\site-packages (from starlette>=0.46.0->fastapi)
(4.12.1)
Requirement already satisfied: idna>=2.8 in .\.venv\Lib\site-packages (from anyio<5,>=3.6.2->starlette>=0.46.0
->fastapi) (3.11)
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO:     Will watch for changes in these directories: ['C:\\Users\\sarik\\Downloads\\fastapi-lab']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [12616] using StatReload

> OUTLINE
> TIMELINE

main.py
Describe what to build
Auto
Local   Default Approvals
```

```
api) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.5 in .\.venv\Lib\site-packages (from pydantic>=2.7.0->fasta
pi) (2.41.5)
Requirement already satisfied: anyio<5,>=3.6.2 in .\.venv\Lib\site-packages (from starlette>=0.46.0->fastapi)
(4.12.1)
Requirement already satisfied: idna>=2.8 in .\.venv\Lib\site-packages (from anyio<5,>=3.6.2->starlette>=0.46.0
->fastapi) (3.11)
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO:     Will watch for changes in these directories: ['C:\\Users\\sarik\\Downloads\\fastapi-lab']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [12616] using StatReload
INFO:     Started server process [16640]
INFO:     Waiting for application startup.
INFO:     Application startup complete.

> OUTLINE
> TIMELINE

main.py
Describe what to build
Auto
Local   Default Approvals

0 0   BLACKBOX Agent   Open Website   Ln 28, Col 30   Spaces: 4   UTF-8   CRLF   {} Python   3.14.2 (.venv)   .venv (3.14.2)   Go Live   BLACKBOXAI: Open Chat   Augment
```

POST /items Add Item

Parameters

Cancel Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute

Responses

Code Description Links

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Pen",
  "price": 20
}'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code Details

200

Response body

```
{
  "message": "Item created",
  "item": {
    "name": "Pen",
    "price": 20
  }
}
```

Download

Response headers

```
content-length: 59
content-type: application/json
date: Tue,17 Mar 2026 15:27:29 GMT
server: uvicorn
```

Responses

Code Description Links

200 Successful Response No links

GET /items Get Items

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code Details

200

Response body

```
[
  {
    "name": "Pen",
    "price": 20
  },
  {
    "name": "Pen",
    "price": 20
  }
]
```

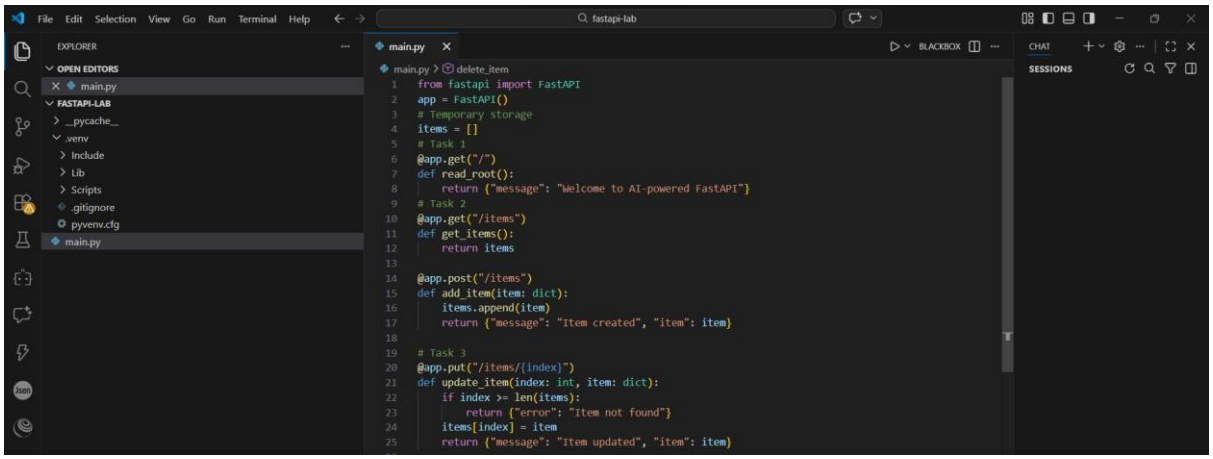
Download

Explanation: GET retrieves all items stored in the list. POST adds a new item into the list.

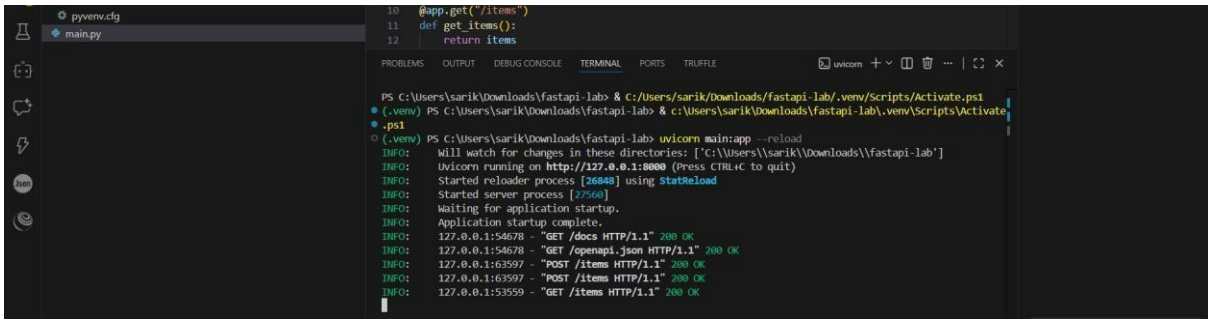
Task Descriptor- 3: (Update Existing Item)

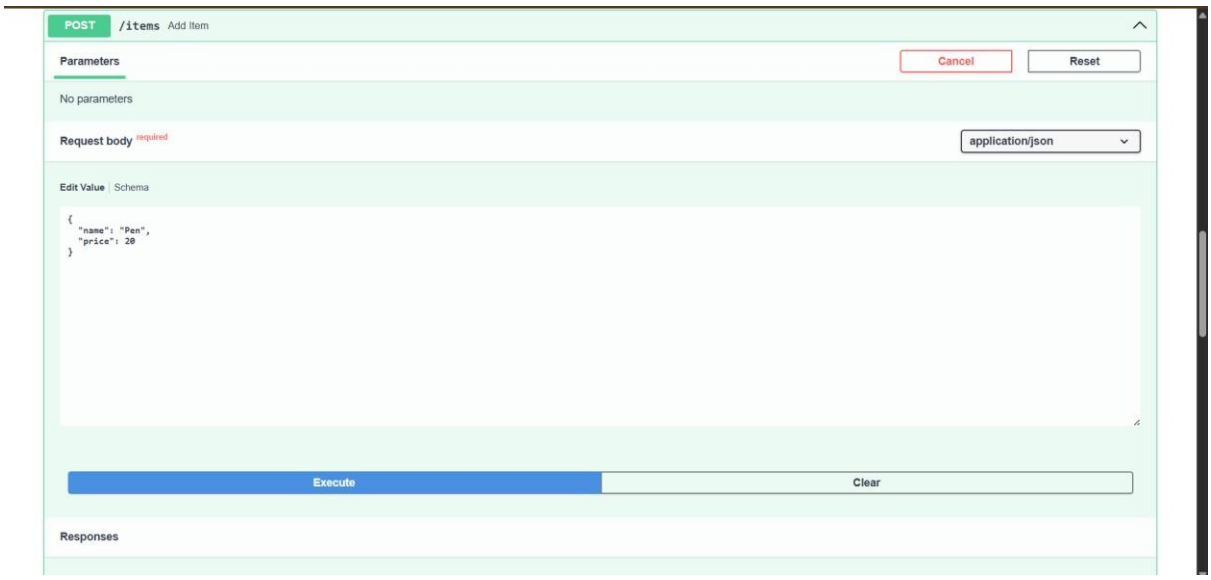
PROMPT: Create a PUT endpoint to update an item using its index.

INPUT:



OUTPUT:





GET /items Get Items

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code Details

200

Response body

```
[
  {
    "name": "Pen",
    "price": 20
  },
  {
    "name": "Pen",
    "price": 20
  }
]
```

Download

PUT /items/{index} Update Item

Parameters

Cancel Reset

Name	Description
index * required integer (path)	0

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pencil",
  "price": 10
}
```

Execute

Execute Clear

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/items/0' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Pencil",
  "price": 10
}'
```

Request URL

```
http://127.0.0.1:8000/items/0
```

Server response

Code Details

200

Response body

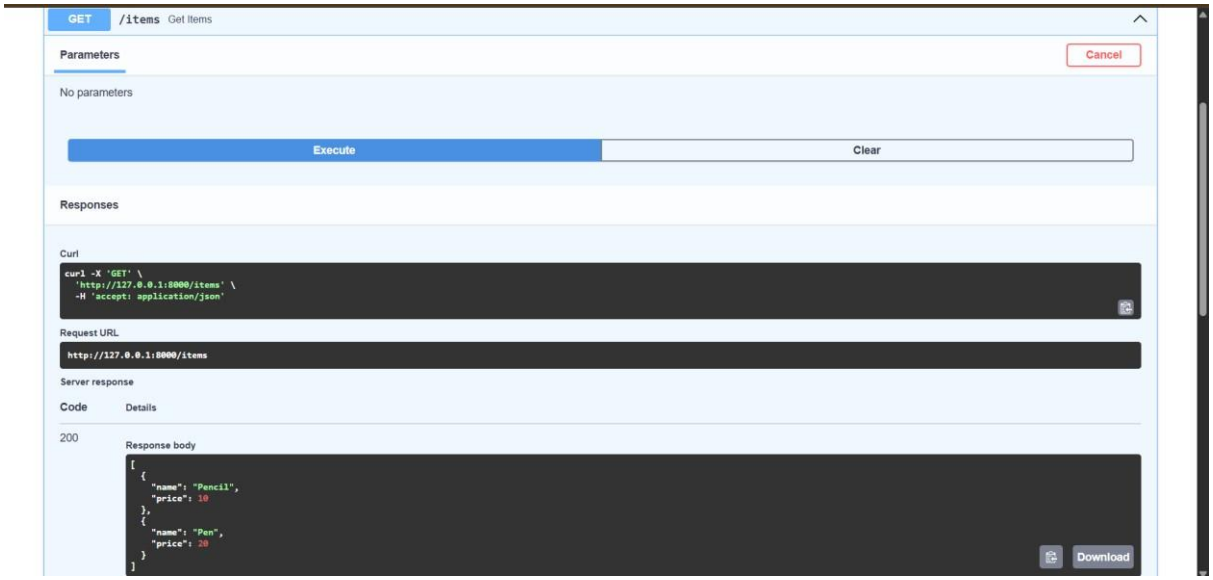
```
{
  "message": "Item updated",
  "item": {
    "name": "Pencil",
    "price": 10
  }
}
```

Download

Response headers

```
content-length: 62
content-type: application/json
date: Tue,17 Mar 2026 15:56:10 GMT
server: uvicorn
```

Responses

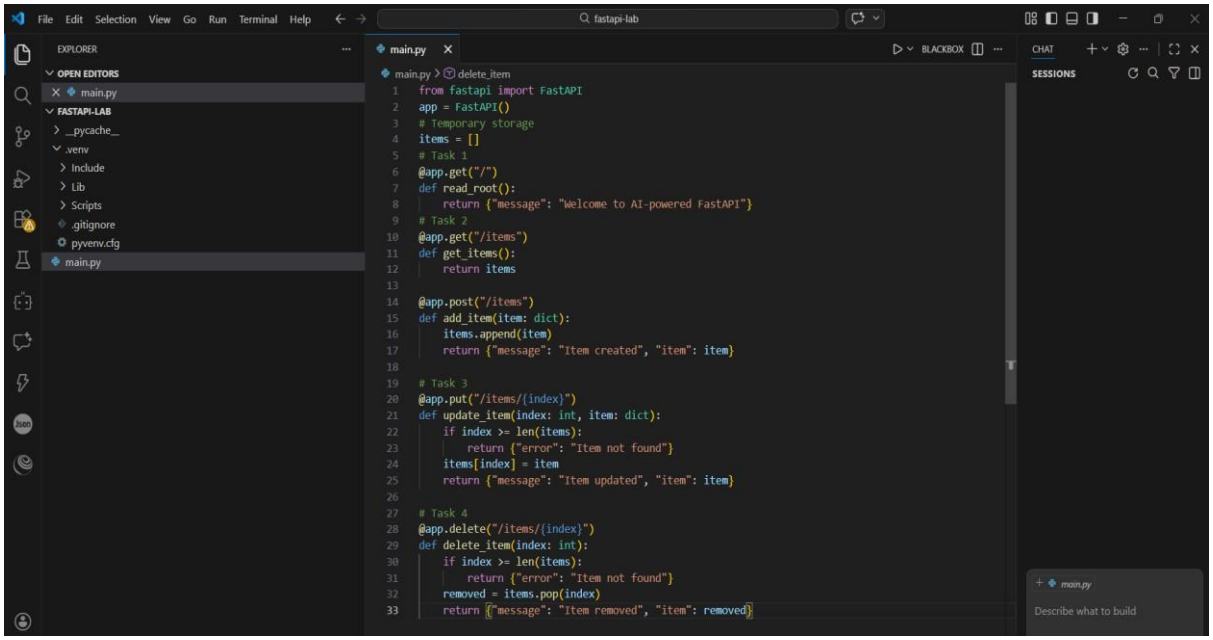


Explanation: PUT modifies an existing item in the list. It checks whether the given index exists.

Task Descriptor- 4: (Delete Item)

PROMPT: Create a DELETE endpoint to remove an item using its index.

INPUT:



OUTPUT:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS TRUFFLE

```
PS C:\Users\sarik\Downloads\fastapi-lab> & C:/Users/sarik/Downloads/fastapi-lab/.venv/Scripts
/Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO:     Will watch for changes in these directories: ['C:\\Users\\sarik\\Downloads\\fastapi
-lab']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [12756] using StatReload
INFO:     Started server process [21644]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Shutting down
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [21644]
INFO:     Stopping reloader process [12756]
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab>
```

uvicorn
uvicorn
powershell

main.py

Describe what to build

POST /items Add Item

Parameters

Cancel Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute Clear

Responses

GET /items Get Items

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

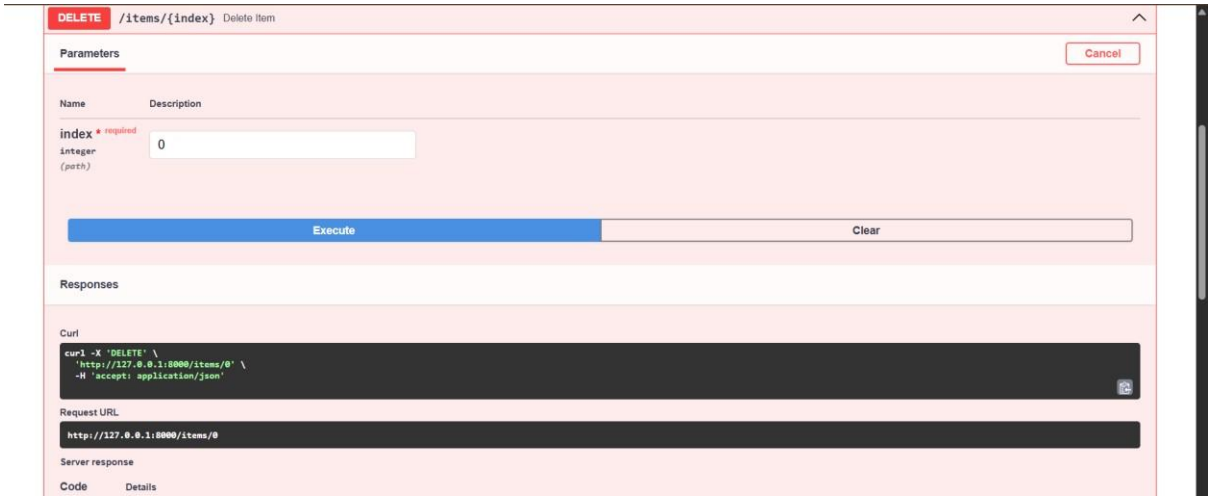
Request URL

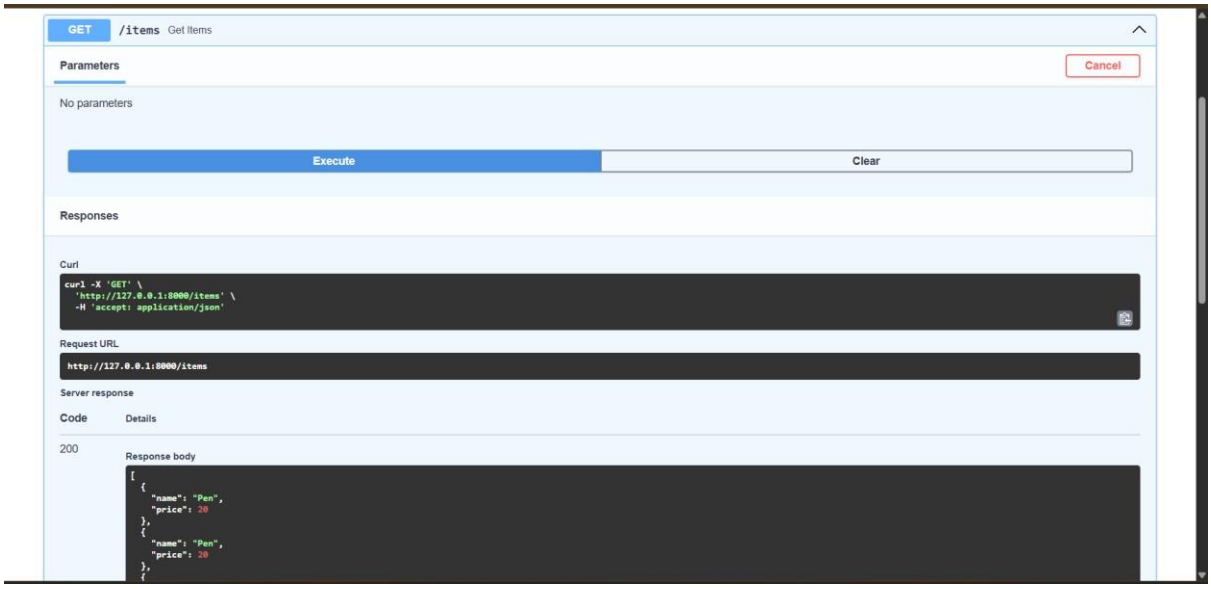
http://127.0.0.1:8000/items

Server response

Code	Details
200	Response body

```
[
  {
    "name": "Pencil",
    "price": 10
  },
  {
    "name": "Pen",
    "price": 20
  },
  {
    "name": "Pen"
```





Explanation: DELETE removes an item from the list. It returns the deleted item as response.

Task Descriptor -5: (Add Auto-Generated Documentation)

PROMPT: Add docstrings to endpoints for automatic Swagger documentation.

INPUT:

```python
from fastapi import FastAPI
app = FastAPI()
# Temporary storage
items = []
# Task 1
@app.get("/")
def read_root():
    """
    GET /
    Returns a welcome message.
    """
    return {"message": "Welcome to AI-powered FastAPI"}
# Task 2
@app.get("/items")
def get_items():
    """
    GET /items
    Returns all items in the list.
    """
    return items
@app.post("/items")
def add_item(item: dict):
    """
    POST /items
    Adds a new item to the list.
    """
    items.append(item)
    return {"message": "Item created", "item": item}
# Task 3
@app.put("/items/{index}")
def update_item(index: int, item: dict):
    """
    PUT /items/{index}
    Updates an item using its index.
    """
    if index >= len(items):
        return {"error": "Item not found"}
```

```python
    if index >= len(items):
        return {"error": "Item not found"}
    items[index] = item
    return {"message": "Item updated", "item": item}
# Task 4
@app.delete("/items/{index}")
def delete_item(index: int):
    """
    DELETE /items/{index}
    Deletes an item using its index.
    """
    if index >= len(items):
        return {"error": "Item not found"}
    removed = items.pop(index)
    return {"message": "Item removed", "item": removed}
```

OUTPUT:

```python
@app.put("/items/{index}")
def update_item(index: int, item: dict):
    """
    PUT /items/{index}
```

```
PS C:\Users\sarik\Downloads\fastapi-lab> & C:/Users/sarik/Downloads/fastapi-lab/.venv/Scripts/Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO:     Will watch for changes in these directories: ['C:\\Users\\sarik\\Downloads\\fastapi-lab']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [10012] using StatReload
INFO:     Started server process [9704]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Shutting down
INFO:     Waiting for application shutdown.
INFO:     Application shutdown complete.
INFO:     Finished server process [9704]
INFO:     Stopping reloader process [10012]
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab>
```

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

GET / Read Root

GET /items Get Items

POST /items Add Item

PUT /items/{index} Update Item

DELETE /items/{index} Delete Item

Schemas

HTTPValidationError > Expand all object

ValidationError > Expand all object

default

GET / Read Root

Parameters

Cancel

No parameters

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body

```
{
  "message": "Welcome to AI-powered FastAPI"
}
```

Download

Response headers


```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body

```
{
  "message": "Welcome to AI-powered FastAPI"
}
```

Download

Response headers


```
content-length: 43
content-type: application/json
date: Tue,17 Mar 2026 15:24:45 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

GET /items Get Items

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code Details

200

Response body

```
[]
```

Download

Response headers

```
content-length: 2
content-type: application/json
date: Tue,17 Mar 2026 15:25:59 GMT
```

POST /items Add Item

Parameters

Cancel Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute

Responses

Code Description Links

GET /items Get Items

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code Details

200

Response body

```
[
  {
    "name": "Pen",
    "price": 20
  },
  {
    "name": "Pen",
    "price": 20
  }
]
```

Download

PUT /items/{index} Update Item

Parameters

Cancel Reset

Name	Description
index * required integer (path)	0

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pencil",
  "price": 10
}
```

Execute

GET /items Get Items

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

Code	Details
200	Response body

```
[
  {
    "name": "Pencil",
    "price": 10
  },
  {
    "name": "Pen",
    "price": 20
  }
]
```

Download

DELETE /items/{index} Delete Item

Parameters

Cancel

Name	Description
index * required integer (path)	0

Execute Clear

Responses

Curl

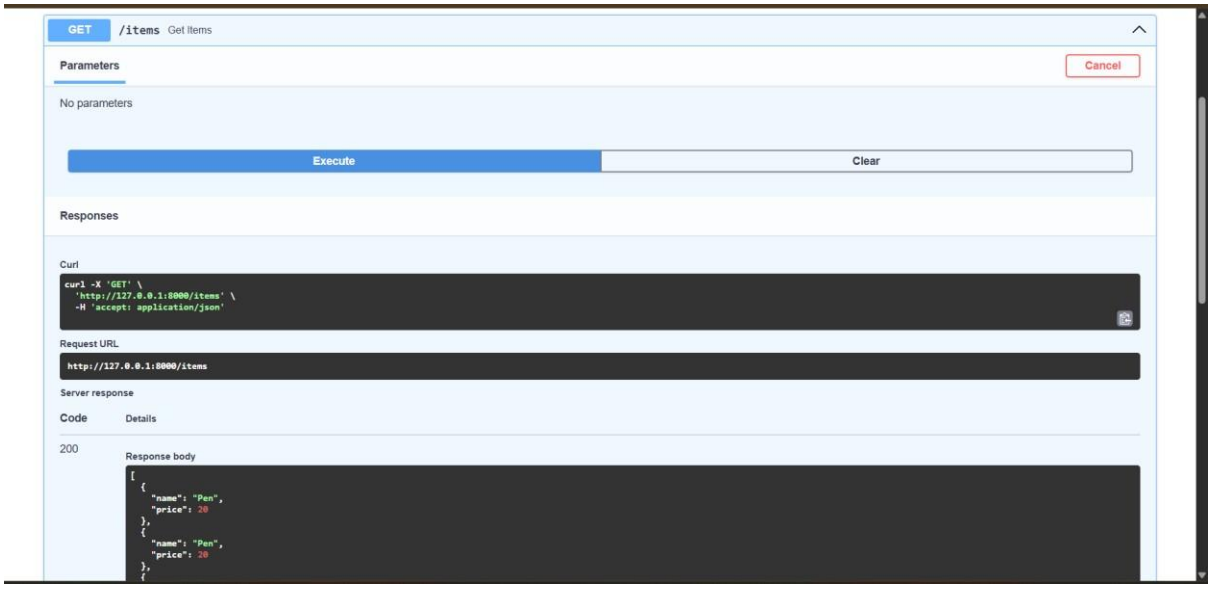
```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/items/0' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items/0
```

Server response

Code	Details



Explanation: FastAPI uses docstrings to generate API documentation. Swagger UI is available at /docs.